Tech / AI Advancements for the Nature Around Us

How Technology can improve our interactions with Nature

Peter Sody - BarCamp Philly 17

About Me: The Background Leading to Tech/AI for Nature

🇩🇪 GERMANY

- Growing up on a farm / winery.
- M.S., Computer Science.
- Robotics: Robug 3.
- AI Startup using constraint solver for staff scheduling.


Robug 3

🇺🇸 UNITED STATES OF AMERICA

- Software Engineer and Architect with a focus on software quality and DevOps.
- Principal Software Engineer: Leading initiatives spanning software quality, security, productivity, and AI adoption for 100 engineers / 15 teams supporting ca. 50 products.
- Gardening / growing vegetables.
- "Junior Arborist".


Junior Arborist

Residential vs. Commercial Applications

🏡 THE RESIDENTIAL DOMAIN

Our Intimate Everyday Nature

- **Immediate Proximity:** Most of of our personal interactions with biological life happen inside living rooms, patios, suburban yards, and domestic pets.
- **Accessible Bio-Intelligence:** Computer vision and smartphone microphones bring enterprise-grade botany and ethology to every household.
- **Rewilding Suburbia:** Collective residential yards represent millions of acres of potential pollinator corridors and micro-habitats.

🏢 COMMERCIAL BENCHMARK

Scale & Macro Impact

- **Broadacre Fields:** Thousands of hectares demanding heavy autonomous tractors and satellite spectral passes.
- **Industrial Supply Chains:** Economic efficiency, bulk food production, and planetary carbon/water footprints.
- **The Technology Transfer:** Commercial agricultural tech (lasers, sensors, CV) is influencing consumer home gadgets.

The 5-Stage Life Cycle at a Glance

Stage 1

🔭 Plan & Observe

Res: Plant CV & backyard audio

Com: Satellites & soil DNA

Stage 2

🌱 Prepare & Plant

Res: Smart seeders & native kits

Com: RTK-GPS tractors & drones

Stage 3

✂️ Weed & Prune

Res: Vision mowers & leaf alerts

Com: CO_2 laser weeders

Stage 4

💧 Feed & Water

Res: Weather ET timers & pot probes

Com: Variable rate drone VRT

Stage 5

🧺 Harvest

Res: Peak kitchen alerts

Com: Soft robotic pickers

Tracing the flow from initial observation to the harvest and regeneration cycle.

Phase 1: Plan / Observe

Establishing baseline ecological intelligence.

Phase 1: Plan / Observe

RESIDENTIAL APPLICATIONS

Houseplants, Yards & Backyard Bio-Listening

- **Smartphone Botanical Diagnostics:** Apps like PlantNet and PictureThis running CNNs on edge devices for immediate leaf pathogen & care advice.
- **Backyard Bioacoustics:** Cornell's *Merlin Bird ID* decoding real-time bird song in home gardens; passive backyard AudioMoth monitoring insect/bat activity.
- **Domestic Companion Ethology:** Smart pet collars (PPG biometric sensors) and acoustic feline/canine vocalization decoders identifying anxiety or distress.
- **Algorithmic Garden Planning:** AI layout apps matching companion plants (tomatoes + basil).

COMMERCIAL BENCHMARK

Planetary & Municipal Scale

- **Hyperspectral Satellites:** Using satellite data for larger ecosystems as well as farming.
- **City Trees:** Municipal AI identifying urban heat islands and street tree canopies via satellite and steet level images.

Phase 1 Spotlight: Real-World In-Action

Citizen Science & Urban Wildlife: Bird Safe Philly via iNaturalist

CITIZEN SCIENCE & MOBILE OBSERVATION

Report a Collision | Bird Safe Philly

- **Crowdsourced Ecological Data:** Everyday citizens in residential and urban areas log bird-window collisions directly into the global **iNaturalist** platform.
- **Geospatial Threat Mapping:** AI and community verification identify species, migratory timing, and deadly glass reflection hotspots across neighborhoods.
- **Direct Conservation Impact:** Data collected fuels the "Lights Out Philly" initiative, prompting architectural retrofits, bird-safe glass installations, and local policy change.



Phase 2: Prepare / Plant

Autonomous Seeding Tractors vs. Domestic Micro-Nurturing.

Phase 2: Prepare / Plant

RESIDENTIAL APPLICATIONS

Kitchen Hydroponics, Seed Pods & Suburban Rewilding

- **Smart Indoor Germination:** Domestic vertical towers (Rise Gardens, Click & Grow) automating LED spectrum ratios (high blue for root emergence, red for foliage).
- **Hyperlocal Native Rewilding:** Citizen apps recommending native wildflower seeds by exact zip-code to establish pollinator-friendly suburban garden patches.
- **Microclimate Soil Prepping:** Bluetooth soil probes verifying substrate temperature and drainage before outdoor transplanting.

COMMERCIAL BENCHMARK

Autonomous Field Robotics

- **RTK-GPS No-Till Drillers:** Autonomous tractors inserting seed at millimetric depths with zero human steering.
- **Robotic Nursery Transplanters:** High-speed optical pick-and-place robots sorting commercial greenhouse plugs.

Phase 2 Spotlight: Real-World In-Action

Autonomous Garden CNC Seeding: FarmBot

PRECISION BACKYARD & COMMUNITY AUTOMATION

FarmBot | Open-Source CNC Farming

- **Precision Cartesian Gantry:** An open-source, automated farming machine designed for residential raised beds, schools, and rooftop gardens.
- **Algorithmic Drag-and-Drop Planting:** Users design polyculture garden layouts graphically in a web app; FarmBot automatically calculates spacing and soil needs.
- **Millimetric Vacuum Seeding:** FarmBot's universal tool head picks specific seeds from onboard bays and injects them at the exact programmed depth for optimal germination.

KEY FEATURES

Tech Specs & Impact

- **Autonomous Seeding:** Zero-guesswork multi-crop planting in home beds.
- **100% Open Source:** Open hardware and open-source Python/React software stack.
- **Project Link:** farm.bot

Phase 3: Weed / Trim / Prune

Industrial Laser Weeders vs. Consumer Vision Robotics.

Phase 3: Weed / Trim / Prune

RESIDENTIAL APPLICATIONS

Vision Robotic Mowers & Handheld Plant Care

- **Boundary-Wire-Free Robotic Mowers:** Consumer units (Mammotion, Segway Navimow) using RTK-GPS + AI stereoscopic cameras to navigate complex lawns without buried wires.
- **Small Animal & Pet Safety AI:** Onboard neural networks halting mower blades instantly upon detecting hedgehogs, ground nesting birds, or family pets.
- **Selective Micro-Trimming:** Vision algorithms distinguishing invasive crabgrass from beneficial clover in home lawns.
- **Houseplant Leaf Pruning Prompts:** Computer vision flagging pest infestations, suggesting selective domestic pruning cuts.

COMMERCIAL BENCHMARK

High-Throughput Destruction

- **Carbon Robotics LaserWeeder:** 30 high-power CO_2 lasers vaporizing 5,000 weeds/min with zero herbicide drift.
- **John Deere See & Spray:** 120-foot boom sprayers micro-dosing weeds at 15 mph.
- **Robotic Vineyard Pruning:** Autonomous robotic arms thinning cane branches based on 3D models.

Phase 3 Spotlight: Real-World In-Action

Autonomous Lawns & Chemical-Free Weeding: BESTMOW & Tertill

AUTONOMOUS LAWN CARE & CUTTING BACK

Robotic lawn mowers

- **BESTMOW:**Philadelphia based robotic lawn mower - developed in collaboration with UPenn
- **Gas to electric:**Accelerates development of robotic / autonomous systems.
- **Enhanced capabilities:**Edging; leaf/snow blowing

CHEMICAL-FREE GARDEN WEEDING

Tertill — Solar Weeding Robot

- **Continuous Solar-Powered Care:** Invented by Joe Jones (co-inventor of Roomba); lives 24/7 inside home garden beds.
- **Height-Based Mechanical Trimming:** Uses capacitive sensors to identify weeds under 1 inch, chopping them down with spinning nylon string before they root.
- **100% Organic & Chemical-Free:** Keeps residential vegetable gardens completely weed-free without pulling or herbicides.

Phase 4: Feed / Fertilize / Water

Delivering exact hydration and nutrition dynamically at the root zone.

Phase 4: Feed / Fertilize / Water

RESIDENTIAL APPLICATIONS

Evapotranspiration AI & Closed-Loop Indoor Hydroponics

- **Weather-Predictive Irrigation:** Smart controllers (Rachio, Eve Aqua) pulling real-time rainfall, solar irradiance, and wind data to prevent over-watering by 30–50%.
- **In-Pot Dielectric Sensors:** Wireless Zigbee/Bluetooth probes in potted plants and raised beds monitoring volumetric water content and fertilizer salinity (EC).
- **Closed-Loop Domestic Hydroponics:** Automated peristaltic nutrient dispensing maintaining perfect EC/pH balance for kitchen greens.
- **Smart Rain Barrel Diversion:** IoT valves redirecting roof stormwater directly to thirsty raised beds during dry spells.

COMMERCIAL BENCHMARK

Broadacre Precision Ag

- **Drone Variable Rate Nutrition:** Multispectral aerial scans directing fertilizer spreaders to apply nitrogen only where needed.
- **Watershed Protection:** Minimizing nitrogen runoff to eliminate algal blooms in municipal water supplies.
- **Canopy Sap-Flow Wearables:** Sensors strapped to commercial almond trees monitoring sap velocity.

Phase 4 Spotlight: Real-World In-Action

Root-Zone Sensor Intelligence: UgMO Technologies at Longwood Gardens

IN-SITU SUBSURFACE SOIL SENSING

Longwood Gardens & UgMO Technologies (KoP)

- **Historic Horticultural Scale:** Longwood Gardens (1,077+ acres of world-renowned display gardens & conservatories) deployed UgMO's underground sensor network.
- **Real-Time Root-Zone Telemetry:** Wireless sensors buried directly at root depths measure soil moisture, temperature, and salinity (fertilizer salts) every 10 minutes.
- **Closed-Loop Smart Irrigation:** Integrates directly with central irrigation controllers, feeding and watering plants only when root-level conditions dictate.
- **Conservation Results:** Reduced water consumption by

KEY TAKEAWAYS

Botanical Precision

- **No Guesswork:** Eliminates schedule-based over-watering and nutrient leaching.
- **Scalable:** Tech applies from botanical conservatories to residential estate lawns.
- **Case Study:** ugmo.com

Phase 5: Harvest

Reaping peak nutritional value and closing the circular organic loop.

Phase 5: Harvest

RESIDENTIAL APPLICATIONS

Vine-Ripe Alerts, Kitchen Freshness & Composting

- **"Cut-and-Come-Again" Guidance:** Indoor hydroponic apps directing growers which outer leaves to harvest first to prolong plant longevity.
- **Circular Soil Loops:** Kitchen composters (Lomi) and yard mulch recycling post-harvest organic scraps into biological soil armor.

COMMERCIAL BENCHMARK

Volume Harvesting & Post-Harvest Preservation

- **Soft-Robotic Field Pickers:** Silicone pneumatic grippers picking delicate berries and tomatoes 24/7 under LED lighting.
- **Biomimetic Protective Coatings:** Apeel Sciences plant-lipid barriers slowing moisture loss and oxidation, eliminating single-use plastic shrink wraps.
- **NIR Spectroscopy:** Non-destructive inline infrared sensors measuring brix sweetness and acidity on conveyor lines.

Phase 5 Spotlight: Real-World In-Action

Autonomous Collaborative Harvest Mobility: Burro

COLLABORATIVE FIELD & FARM HARVESTING

Burro — Autonomous Field Mobility (Philadelphia)

- **Vision-Based Autonomous Assistant:** An autonomous mobile robot (AMR) using computer vision and 360° cameras to navigate outdoor crop rows alongside human pickers without GPS or buried wires.
- **"Follow-Me" Harvest Carrying:** Follows harvesting crews through table-grape vineyards, blueberry orchards, and nursery rows, carrying up to 500 lbs of freshly picked produce directly to collection trailers.
- **Doubling Harvest Efficiency:** Eliminates miles of backbreaking manual fruit-carrying, boosting human harvesting productivity by up to 100%.

KEY FEATURES

The Burro Platform

- **Modular Robotics:** Evolves from harvest carrier into scouting, towing, and micro-spraying platform.
- **Commercial Fleet:** Hundreds of thousands of autonomous miles logged in active farms.
- **Company Link:** burro.ai

Harvesting in Action: Human-in-the-loop collaborative robotics (Burro) solving labor-intensive crop transport directly

Synthesis: Residential vs. Commercial Across the 5 Phases

Life Cycle Phase	🏡 Residential Focus (Home, Garden & Pets)	🏙️ Commercial Scale (Farms & Forests)
1. Plan / Observe	Smartphone botany (PlantNet), backyard audio (Merlin), pet health telemetry	Satellite, drone, street level imaging
2. Prepare / Plant	Kitchen hydroponic towers, native pollinator seed kits, smart potting media	RTK-GPS no-till drillers
3. Weed & Prune	Vision robotic mowers with hedgehog safety, houseplant pruning alerts	5,000 shot/min CO_2 laser weeders, See & Spray booms, 3D vineyard arms
4. Feed & Water	Evapotranspiration smart garden timers, in-pot wireless capacitive probes	Multispectral drone fertilizer prescriptions, sap sensors
5. Harvest & Store	Vine-ripeness vision prompts, kitchen compost, Apeel shelf-life extension	Soft-robotic fruit pickers, Apeel protective plant barrier

Practical Toolkit: Bringing Nature AI Home Today

FREE & IMMEDIATE

📱 Smartphone Bio-Sensors

- **Merlin Bird ID:** Acoustic bird song classification in your yard.
- **PlantNet / PictureThis:** Instant diagnosis of pests and leaf chlorosis.
- **iNaturalist:** Mapping backyard native insects and pollinators.

UNDER $100

🌱 Ambient Domestic IoT

- **Smart Soil Probes:** Bluetooth capacitive moisture & nutrient sensors for potted plants.
- **Smart Hose Timers:** Evapotranspiration weather-predictive watering controllers.
- **Indoor Plant LEDs:** Full-spectrum smart lighting on circadian timers.

HOME UPGRADES

🤖 Autonomous Stewardship

- **Vision Robotic Mowers:** Wire-free lawn maintenance protecting backyard wildlife.
- **Kitchen Hydroponic Towers:** Fresh daily greens with automated nutrient dosing.
- **Smart Composters:** Turning kitchen organic waste into rich soil food.

References & Further Reading (Residential & Commercial)

🏡 Residential Technologies & Studies

- **Cornell Lab of Ornithology (BirdNET / Merlin):** *Deep learning for acoustic avian diversity monitoring in residential areas.* (birdnet.cornell.edu)
- **FarmBot:** *Open-Source CNC Farming*

🏢 Commercial Field Evaluations

- **Carbon Robotics:** *LaserWeeder commercial field evaluations (CO_2 lasers & 90% herbicide reduction).* (carbonrobotics.com)
- **Apeel Sciences:** *Plant-derived edible coatings slowing transpiration and oxidation to combat domestic produce waste* (apeel.com).

Thank You!

Questions & Discussion

🌿 **Discussion:**

- How can technology further reconnect modern households with nature?
- To what degree do we want to utilize tech in nature?
- Is current tech leading us to what we want for nature?

📫 **Contact:** nature@sody.us

💻 **Slide Deck:** github: pesody/tech-nature